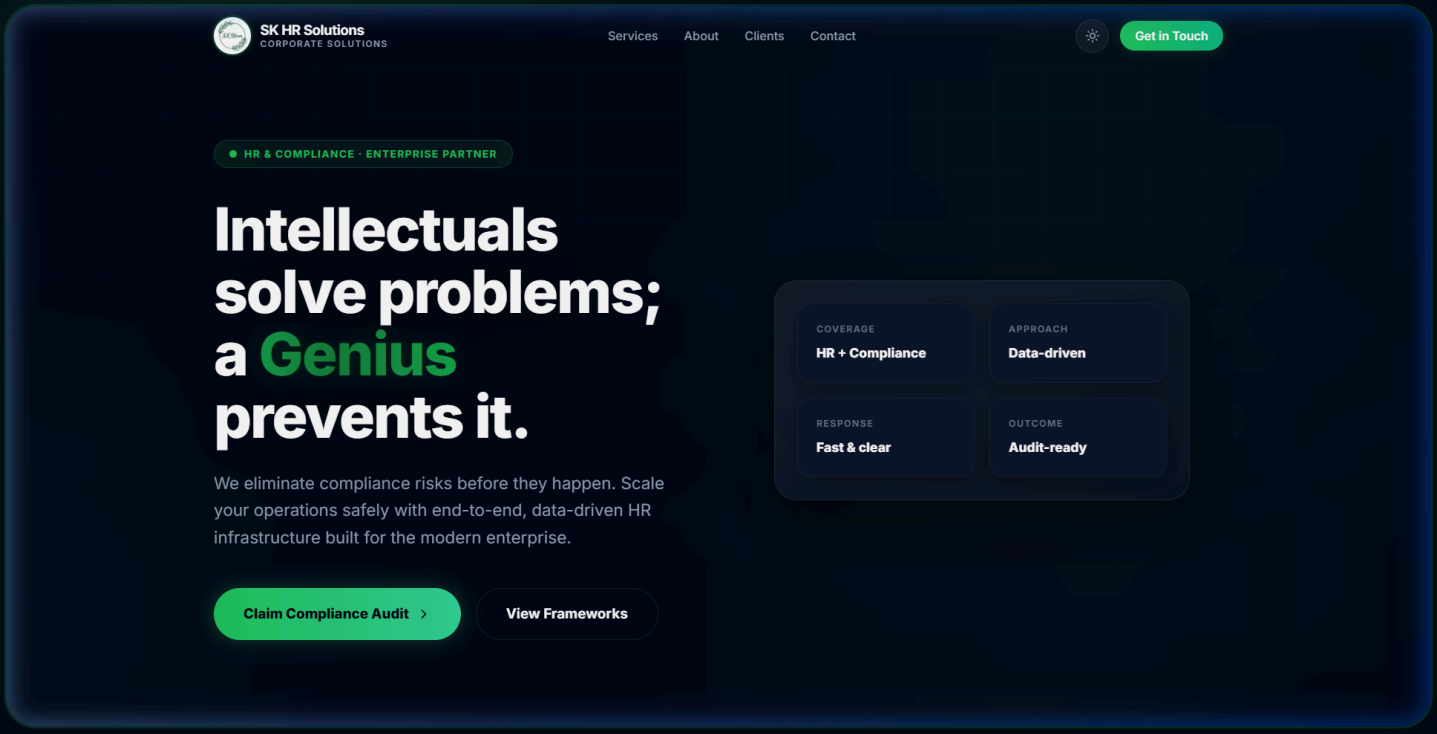
SK HR Solutions Platform Report

ENTERPRISE SAAS ARCHITECTURE & DEPLOYMENT



Executive Summary

The SK HR Solutions platform is a fully-scalable, high-performance web application designed for a premium user experience and bulletproof data security. By leveraging modern headless architecture, the system cleanly isolates the high-fidelity Next.js frontend from the robust Node.js backend infrastructure, ensuring maximum uptime, SEO superiority, and military-grade performance.

Core Technology Stack

1. FRONTEND ECOSYSTEM (THE INTERFACE)

- Next.js 14 (App Router)
- React 18
- TypeScript
- Tailwind CSS v3
- Framer Motion (3D & Animations)

2. BACKEND INFRASTRUCTURE (THE ENGINE)

- Node.js
- Express.js (REST API)
- TypeScript
- Bcrypt (Cryptography)
- JWT Auth

3. DATABASE & STATE (THE MEMORY)

- MongoDB Atlas (Cloud NoSQL)
- Mongoose ODM
- SWR (Client Data Fetching)

4. DEVSECOPS & DEPLOYMENT (THE CLOUD)

- Netlify Edge (Frontend CDN)
- Render Cloud (Backend Host)
- Nodemailer (SMTP Relay)
- GitHub (VCS)

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System Capabilities & Security

HIGHLIGHTING THE PREMIUM FEATURES OF THE PLATFORM

Premium User Experience (UX/UI)

Hardware-Accelerated Animations

Utilizing Framer Motion to deliver 60fps micro-interactions without jitter.

- ✓ Staggered entrance reveals
- ✓ Magnetic mouse-tracking buttons
- ✓ Scroll-progress indicators (shimmer)
- ✓ 3D tilting cards based on cursor bounds

Glassmorphic Design System

A sophisticated blend of frosted-glass overlays and dark-mode elegance.

- ✓ Adaptive CSS backdrop-filters
- ✓ Gradient mesh ambient backgrounds
- ✓ Smooth Theme Toggle (Light/Dark)
- ✓ Intelligent navbar scroll-hiding

Security & Authentication Hardening

Brute-Force Protection

The admin gateway is strictly fortified against malicious intrusion.

- ✓ express-rate-limit active
- ✓ 15-min IP lockouts after 5 bad tries
- ✓ Intentional latency simulation on auth

Secure Session Management

Zero-trust methodology for handling administrative tokens across boundaries.

- ✓ Bcrypt password hashing (Salt Rounds: 10)
- ✓ Strict cross-origin CSRF defenses
- ✓ Cookie-based JWTs (SameSite=None, Secure)

Core Features & Pipelines

Headless CMS Engine

Custom MongoDB pipeline powering the dynamic Testimonials, Clients, and Services blocks without requiring hard-code changes.

Async Email Relays

Nodemailer SMTP integration instantly fires rich HTML inquiry alerts to the owner, executed asynchronously for zero UI latency.

Global Edge Delivery

Deployed strictly on Tier-1 global CDNs (Netlify) ensuring sub-second load times regardless of client geography.